

TOI-1516b

R-2107

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI1516_250830 · created 2026-08-10T23:54:51+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460917.903 +/- 0.048 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.115 +/- 0.043
Transit depth (Rp/Rs)^2	1.33 +/- 0.99 [%]
Orbital Inclination (inc)	89.86 +/- 1.43
Airmass coefficient 1 (a1)	1.303 +/- 0.027
Airmass coefficient 2 (a2)	-0.182 +/- 0.048
Scatter in the residuals of the lightcurve fit is	2.04 %
Best Comparison Star	#2 - [435, 321]
Optimal Aperture	2.96
Optimal Annulus	5.7
Transit Duration (day)	0.07 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.115 ± 0.043 vs 0.1224 (deviation 0.17σ)
Duration fit vs published	0.07 d vs 0.1217 d (deviation 1.52σ)
Mid-time O–C	-99.6 min
Depth SNR	2.7
Residual scatter	2.04 %

Light curve

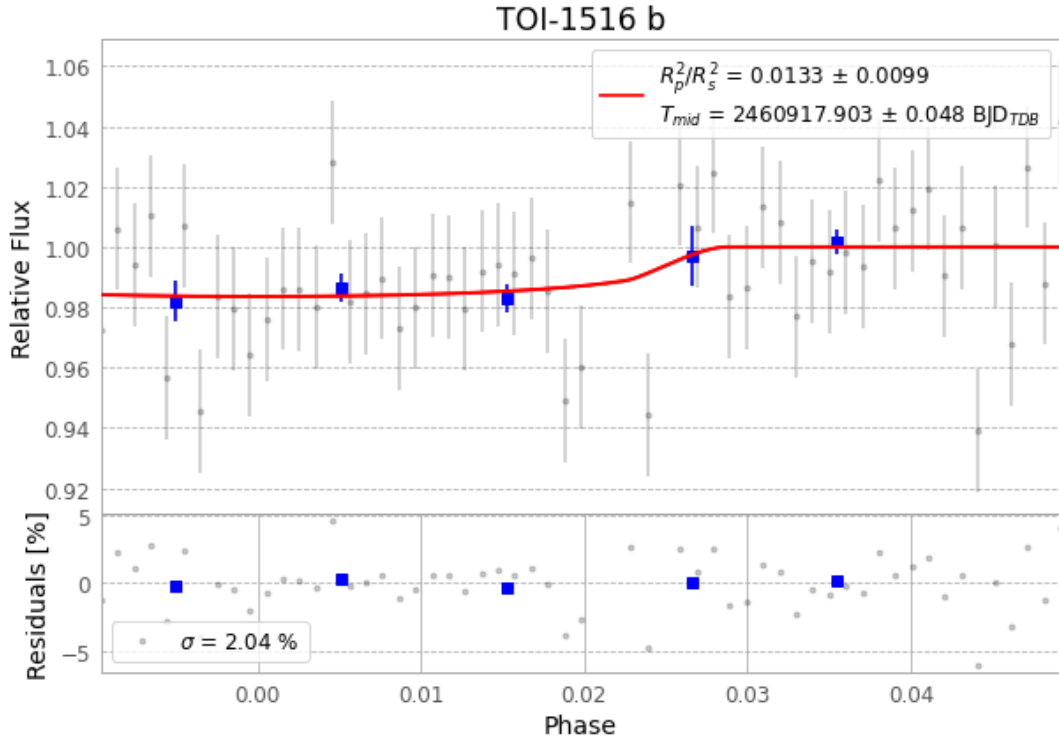


Figure 1 — FinalLightCurve_TOI-1516 b_2025-08-30.png (SHA-256 a82cb767aff8f30d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [355, 263], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 80d4d1422d087cea...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

59 FITS frames (39 MB), TOI1516250830090915.FITS → TOI1516250830120315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI1516-250830.log (3b2c1647d082...), FinalLightCurve_TOI-1516 b_2025-08-30.png (a82cb767aff8...), FinalLightCurve_TOI-1516 b_2025-08-30.csv (f6f8101c2810...)

Compute resources

Wall time 156.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 23:54Z.